

■ CAMELOT-OS: MASTER SYSTEM SPECIFICATION

****Version:**** 400.1.0 (Lattice Radiant)
****Classification:**** SOVEREIGN / CONFIDENTIAL
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1. ARCHITECTURAL OVERVIEW: THE SINGULARITY LATTICE

Camelot-OS is a split-topology agentic operating system designed to bifurcate Cloud-based Neural Reasoning from Edge-based Kinetic Execution. This architecture eliminates the "hallucination-to-metal" risk by enforcing a strict ****Septem Regna**** governance layer.

1.1 Structural Visualization (Mermaid)

```
```mermaid
graph TD
 subgraph L7_L3_Neural_Cloud
 L7[L7: ETHEREAL - Anya_Ω] -->|Titan Prompt| L3[L3: NEURAL - Merlin_Ω]
 L3 -->|HTN Task Tree| L5[L5: AGENTIC - Paladin_Ω]
 L6[L6: GOVERNANCE - Arthur_Ω] -. ->|HITL Gate| L5
 end

 subgraph L4_Memory
 L4[L4: SEMANTIC - Chronos_Ω] <-->|UKG Sync| L3
 end

 subgraph L2_L1_Kinetic_Edge
 L5 -->|Map-Reduce| L2[L2: KINETIC - Lukas_Ω]
 L2 -->|Saltare/Cribo| L1[L1: SUBSTRATE - Morgana_Ω]
 L1 -->|Metal/Docker| Hardware
 end
```
```

2. THE SEPTEM REGNA (7 SOVEREIGN LAYERS)

| LAYER | DESIGNATION | PRIMARY ENGINE | CORE FUNCTION |
|--------|--------------|----------------|--|
| --- | --- | --- | --- |
| **L7** | **ETHEREAL** | APEE v6.5 | Intent Compilation & Triple-QFT Renormalization. |

| **L6** | **GOVERNANCE** | Iron Gate v1.1 | Policy Enforcement & Copyright Shield (CES). |
| **L5** | **AGENTIC** | SARDA Engine | Swarm Orchestration & Map-Reduce Dispatch. |
| **L4** | **SEMANTIC** | Ouroboros | Immutable Truth Graphs & UKG JSON-LD Memory. |
| **L3** | **NEURAL** | Videnepus | High-Criticality Reasoning & LaC Oscillation. |
| **L2** | **KINETIC** | Lukas Stack | Zero-Latency Binary Execution (Rust/Go). |
| **L1** | **SUBSTRATE** | Modal/Bridge | Hardware Interface & Sandbox Isolation. |

3. CORE PROTOCOLS

3.1 Triple-QFT (Quantization, Qualification, Transformation)

Processes raw human vibe into machine-readable directives. Strips conversational filler (Physics) and identifies anchor tokens (Engineering).

3.2 NDR+S (Neurosymbolic Deep Reasoning + Synthesis)

Integrates symbolic task trees (HTN) with neural intuition. Enforces the **Trinity Validation Check** (Logic, Risk, Alignment) before any disk commit.

3.3 Bio-Kinetic Swarm Zoology

Deploys 150-token **Nano-Knights** for high-velocity parallel work:

- **Formica (Ants)**: Parallel rewriting.
- **Pongid (Gorillas)**: API/SDK heavy-lifting.
- **Castor (Beavers)**: Security dams and sandboxing.

4. KINETIC PURITY METRICS

- **Baseline Purity**: 98/100.
- **Memory Ceiling**: 8GB (Operational at 1.4GB).
- **Latency**: < 1.2s for full system bootstrap.

[STAMP]: RADIANT LATTICE v400.1.0
[VALIDATED]: By Sir Gideon ■ (Scorpion Pass)